



Parallel and Perpendicular Lines

Grades 9-12 • SAT / ACT mathematics foundations

Algebra • Use slopes to relate lines in a coordinate plane.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Use slopes to relate lines in a coordinate plane.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. A line has slope $\frac{3}{4}$. Write equations for a parallel line and a perpendicular line through $(8, 10)$.

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $\frac{3}{4} \times (-\frac{4}{3}) = -1$. Point-slope form ensures each line passes through $(8,10)$.

Model 2. A line has slope $\frac{4}{5}$. Write equations for a parallel line and a perpendicular line through $(5, 8)$.

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $\frac{4}{5} \times (-\frac{5}{4}) = -1$. Point-slope form ensures each line passes through $(5,8)$.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A line has slope $\frac{2}{3}$. Write equations for a parallel line and a perpendicular line through $(6, 10)$.

2. A line has slope $\frac{2}{9}$. Write equations for a parallel line and a perpendicular line through $(7, 6)$.

3. A line has slope $\frac{5}{3}$. Write equations for a parallel line and a perpendicular line through $(6, 9)$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A line has slope $\frac{4}{3}$. Write equations for a parallel line and a perpendicular line through $(5, 12)$.

5. A line has slope 3. Write equations for a parallel line and a perpendicular line through $(3, 10)$.

6. A line has slope 1. Write equations for a parallel line and a perpendicular line through $(2, 6)$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A line has slope $\frac{3}{4}$. Write equations for a parallel line and a perpendicular line through $(8, 7)$.

8. A line has slope $\frac{2}{7}$. Write equations for a parallel line and a perpendicular line through $(4, 6)$.

9. A line has slope $\frac{7}{6}$. Write equations for a parallel line and a perpendicular line through $(2, 11)$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A line has slope $\frac{3}{2}$. Write equations for a parallel line and a perpendicular line through $(4, 7)$.

11. A line has slope $\frac{5}{7}$. Write equations for a parallel line and a perpendicular line through $(8, 9)$.

12. A line has slope $\frac{4}{9}$. Write equations for a parallel line and a perpendicular line through $(7, 8)$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A line has slope $\frac{3}{4}$. Write equations for a parallel line and a perpendicular line through $(8, 7)$.

14. A line has slope 2. Write equations for a parallel line and a perpendicular line through $(5, 8)$.

15. A line has slope $\frac{9}{8}$. Write equations for a parallel line and a perpendicular line through $(6, 13)$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A line has slope $\frac{4}{3}$. Write equations for a parallel line and a perpendicular line through $(7, 8)$.
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Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A line has slope 3. Write equations for a parallel line and a perpendicular line through (2, 13).

2. A line has slope $\frac{7}{5}$. Write equations for a parallel line and a perpendicular line through (4, 11).

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A line has slope 2. Write equations for a parallel line and a perpendicular line through (6, 10).

4. A line has slope $\frac{5}{2}$. Write equations for a parallel line and a perpendicular line through (6, 9).



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A line has slope 4. Write equations for a parallel line and a perpendicular line through (2, 12).

6. A line has slope $\frac{3}{7}$. Write equations for a parallel line and a perpendicular line through (7, 7).



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A line has slope 2. Write equations for a parallel line and a perpendicular line through (5, 10).

2. A line has slope 1. Write equations for a parallel line and a perpendicular line through (2, 11).



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A line has slope $\frac{8}{5}$. Write equations for a parallel line and a perpendicular line through $(4, 12)$.

4. A line has slope $\frac{3}{2}$. Write equations for a parallel line and a perpendicular line through $(4, 13)$.



Practice worked solutions

1. Parallel: $y - 10 = 2/3(x - 6)$; perpendicular: $y - 10 = -3/2(x - 6)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2/3 \times (-3/2) = -1$.
Point-slope form ensures each line passes through (6,10).

2. Parallel: $y - 6 = 2/9(x - 7)$; perpendicular: $y - 6 = -9/2(x - 7)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2/9 \times (-9/2) = -1$.
Point-slope form ensures each line passes through (7,6).

3. Parallel: $y - 9 = 5/3(x - 6)$; perpendicular: $y - 9 = -3/5(x - 6)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $5/3 \times (-3/5) = -1$.
Point-slope form ensures each line passes through (6,9).

4. Parallel: $y - 12 = 4/3(x - 5)$; perpendicular: $y - 12 = -3/4(x - 5)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $4/3 \times (-3/4) = -1$.
Point-slope form ensures each line passes through (5,12).

Practice worked solutions

5. Parallel: $y - 10 = 3(x - 3)$; perpendicular: $y - 10 = -1/3(x - 3)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3 \times (-1/3) = -1$. Point-slope form ensures each line passes through (3,10).

6. Parallel: $y - 6 = 1(x - 2)$; perpendicular: $y - 6 = -1(x - 2)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $1 \times (-1) = -1$. Point-slope form ensures each line passes through (2,6).

7. Parallel: $y - 7 = 3/4(x - 8)$; perpendicular: $y - 7 = -4/3(x - 8)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3/4 \times (-4/3) = -1$. Point-slope form ensures each line passes through (8,7).

8. Parallel: $y - 6 = 2/7(x - 4)$; perpendicular: $y - 6 = -7/2(x - 4)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2/7 \times (-7/2) = -1$. Point-slope form ensures each line passes through (4,6).

Practice worked solutions

9. Parallel: $y - 11 = 7/6(x - 2)$; perpendicular: $y - 11 = -6/7(x - 2)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $7/6 \times (-6/7) = -1$.
Point-slope form ensures each line passes through (2,11).

10. Parallel: $y - 7 = 3/2(x - 4)$; perpendicular: $y - 7 = -2/3(x - 4)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3/2 \times (-2/3) = -1$.
Point-slope form ensures each line passes through (4,7).

11. Parallel: $y - 9 = 5/7(x - 8)$; perpendicular: $y - 9 = -7/5(x - 8)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $5/7 \times (-7/5) = -1$.
Point-slope form ensures each line passes through (8,9).

12. Parallel: $y - 8 = 4/9(x - 7)$; perpendicular: $y - 8 = -9/4(x - 7)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $4/9 \times (-9/4) = -1$.
Point-slope form ensures each line passes through (7,8).

Practice worked solutions

13. Parallel: $y - 7 = 3/4(x - 8)$; perpendicular: $y - 7 = -4/3(x - 8)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3/4 \times (-4/3) = -1$. Point-slope form ensures each line passes through (8,7).

14. Parallel: $y - 8 = 2(x - 5)$; perpendicular: $y - 8 = -1/2(x - 5)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2 \times (-1/2) = -1$. Point-slope form ensures each line passes through (5,8).

15. Parallel: $y - 13 = 9/8(x - 6)$; perpendicular: $y - 13 = -8/9(x - 6)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $9/8 \times (-8/9) = -1$. Point-slope form ensures each line passes through (6,13).

16. Parallel: $y - 8 = 4/3(x - 7)$; perpendicular: $y - 8 = -3/4(x - 7)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $4/3 \times (-3/4) = -1$. Point-slope form ensures each line passes through (7,8).

Transfer worked solutions

1. Parallel: $y - 13 = 3(x - 2)$; perpendicular: $y - 13 = -1/3(x - 2)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3 \times (-1/3) = -1$. Point-slope form ensures each line passes through (2,13).

2. Parallel: $y - 11 = 7/5(x - 4)$; perpendicular: $y - 11 = -5/7(x - 4)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $7/5 \times (-5/7) = -1$. Point-slope form ensures each line passes through (4,11).

3. Parallel: $y - 10 = 2(x - 6)$; perpendicular: $y - 10 = -1/2(x - 6)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2 \times (-1/2) = -1$. Point-slope form ensures each line passes through (6,10).

4. Parallel: $y - 9 = 5/2(x - 6)$; perpendicular: $y - 9 = -2/5(x - 6)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $5/2 \times (-2/5) = -1$. Point-slope form ensures each line passes through (6,9).

Transfer worked solutions

5. Parallel: $y - 12 = 4(x - 2)$; perpendicular: $y - 12 = -1/4(x - 2)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $4 \times (-1/4) = -1$. Point-slope form ensures each line passes through (2,12).

6. Parallel: $y - 7 = 3/7(x - 7)$; perpendicular: $y - 7 = -7/3(x - 7)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3/7 \times (-7/3) = -1$. Point-slope form ensures each line passes through (7,7).

Exit-ticket worked solutions

1. Parallel: $y - 10 = 2(x - 5)$; perpendicular: $y - 10 = -1/2(x - 5)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $2 \times (-1/2) = -1$. Point-slope form ensures each line passes through (5,10).

2. Parallel: $y - 11 = 1(x - 2)$; perpendicular: $y - 11 = -1(x - 2)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $1 \times (-1) = -1$. Point-slope form ensures each line passes through (2,11).

3. Parallel: $y - 12 = 8/5(x - 4)$; perpendicular: $y - 12 = -5/8(x - 4)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $8/5 \times (-5/8) = -1$. Point-slope form ensures each line passes through (4,12).

4. Parallel: $y - 13 = 3/2(x - 4)$; perpendicular: $y - 13 = -2/3(x - 4)$

Parallel slopes are equal. Perpendicular slopes are negative reciprocals: $3/2 \times (-2/3) = -1$. Point-slope form ensures each line passes through (4,13).

Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Use slopes to relate lines in a coordinate plane. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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